Epics

CareQueue - Hospital Digital Token and Queue Management System

Field	Value
Document	Epics (Product Backlog, part 1 of 2)
Product	CareQueue - Hospital Digital Token and Queue Management System
Problem statement	Patients wait in queues for registration and billing. Build a digital token and queue-management system that displays the current token and estimated waiting time.
Traces to	CareQueue MRD v1.0 and CareQueue PRD v1.0
Version / Status	v1.0 - Draft for hackathon submission
Date	24 October 2026
Team	Future Coders
Roles	BA: Sam Austin Dev: Ashik Rasool UI/UX Designer: Nishandhini QA Tester: Rogasree

1. Overview

The backlog has **6 epics** and **25 user stories**. Each epic traces to a feature in the PRD and a market requirement in the MRD. The three Must-Have features (M1-M3) are covered by 7 Must stories; everything else is Should or Could and is built only after the Musts pass. E5 (sign-in) and E6 (queue reports) were added by the team after the first backlog of 4 epics and 13 stories, US-17 to US-20 (the PRD's Could-Haves C1, C3, C4 and patient feedback) after that, and US-21 to US-25 (features beyond a standard queue) last.

Epic	Priority	PRD features	MRD needs	Stories
E1 Token issue	Must	M1, S4	MR-1, MR-4	US-01, US-02, US-03, US-04, US-17
E2 Calling and Now-Serving display	Must	M2, S1, S2	MR-2	US-05, US-06, US-07, US-08, US-18, US-22
E3 Waiting-time estimate	Must	M3, S5	MR-3	US-09, US-10, US-11, US-21
E4 Patient's token copy	Should / Could	S3	MR-1, MR-3	US-12, US-13, US-19, US-23
E5 Sign-in and access	Should	S6	MR-1, MR-2	US-14
E6 Queue reports	Could	C2	MR-3, MR-4	US-15, US-16, US-20, US-24, US-25

2. Epic Details

E1 - Token issue (Must)

Field	Detail
User need	Every walk-in patient needs a fair, visible place in the queue instead of a handwritten slip or no number at all.
Goal	The receptionist issues a sequential Registration (R) or Billing (B) token in under 15 seconds, and moves a patient from registration to billing without losing their place in the flow.
Traces to	PRD: M1, S4 | MRD: MR-1, MR-4
Stories	US-01 Issue a token for the right queue (Must, M) US-02 Number tokens in order, reset daily (Must, S) US-03 Refuse missing or invalid details (Must, S) US-04 Send a patient on to billing (Should, M) US-17 Call priority patients first (Could, M)
Definition of done	A token can be created for either queue from the staff screen; numbers never repeat within a day and reset the next day; invalid details are refused with a clear message; all Must acceptance criteria pass.

E2 - Calling and Now-Serving display (Must)

Field	Detail
User need	Patients need to see whose turn it is without crowding the counter; the receptionist needs to call the next patient without shouting.
Goal	One click calls the next token, and a full-screen waiting-room display shows the token being served and the next five, updated live.
Traces to	PRD: M2, S1, S2 | MRD: MR-2
Stories	US-05 Call the next patient with one click (Must, S) US-06 Show now-serving on the waiting-room display (Must, M) US-07 Skip or recall an absent patient (Should, S) US-08 Announce the called token (Should, S) US-18 See and hear the queue in Tamil (Could, M) US-22 Hold my place while I step out (Could, M)
Definition of done	Call Next advances the queue; the display updates within 2 seconds without refresh and never shows patient names; skip/recall and the voice call work if built; all Must acceptance criteria pass.

E3 - Waiting-time estimate (Must)

Field	Detail
User need	Patients need to know roughly how long they will wait so they can sit down or step out calmly.
Goal	Every waiting token shows an estimated waiting time that starts from a default and learns from the clinic's real service speed.
Traces to	PRD: M3, S5 | MRD: MR-3
Stories	US-09 Show an estimated wait for every token (Must, M) US-10 Learn from real service times (Must, M) US-11 Serve from more than one counter (Should, L) US-21 Tell patients when the doctor is delayed (Could, M)
Definition of done	EWT = R + N x AST is shown for every waiting token and recalculated on every queue event; it switches to real service times after 3 tokens are served; all Must acceptance criteria pass.

E4 - Patient's token copy (Should / Could)

Field	Detail
User need	Patients want to keep their token with them and check it without watching the screen.
Goal	The patient gets a printable or downloadable slip, and can scan a QR code to follow their token live on their own phone.
Traces to	PRD: S3 | MRD: MR-1, MR-3
Stories	US-12 Print or download the token slip (Could, S) US-13 Track my token on my phone (Should, M) US-19 Get an alert when my turn is near (Could, S) US-23 Share my token with family (Could, S)
Definition of done	The slip prints on one page with token, queue and estimate; the QR page shows the patient's token, the token being served, people ahead and the estimate, live, with no login.

E5 - Sign-in and access (Should)

Field	Detail
User need	Patients and reception staff need to reach their own screen quickly, and the staff console must not be open to patients in the waiting room.
Goal	A landing page lets a patient sign in with name and phone and a receptionist with a PIN; the token page and the staff console each open only for their role.
Traces to	PRD: S6 | MRD: MR-1, MR-2
Stories	US-14 Sign in as patient or receptionist (Should, M)
Definition of done	Both sign-ins work, a wrong PIN is refused, the token page and staff console send anyone not signed in back to the landing page, and all four acceptance criteria pass in browser tests.

E6 - Queue reports (Could)

Field	Detail
User need	The hospital needs to know how the queue really performed today (how many patients, how long they waited, when it was busiest) to plan staff, without counting slips by hand.
Goal	A staff-only Reports page shows today's numbers per department, live: tokens issued, served, waiting and missed, the average wait and service time, a tokens-per-hour chart with the busiest hour, and a CSV download.
Traces to	PRD: C2 | MRD: MR-3, MR-4
Stories	US-15 See today's queue summary (Could, M) US-16 Find the busiest hour and export the day (Could, S) US-20 Rate my visit (Could, S) US-24 Check how accurate the wait estimates are (Could, M) US-25 Find the best time to visit (Could, S)
Definition of done	The numbers match the staff console, update without refresh, never show patient names or phone numbers, the CSV opens in a spreadsheet with the same figures, the page is closed to anyone not signed in as staff, and all acceptance criteria pass in tests.

3. Traceability Chain

Every feature in the demo must trace back: **demo -> story -> epic -> PRD feature -> user need**.

User need (MRD)	PRD feature	Epic	Stories
MR-1 Every patient gets a unique, ordered token	M1 Token generation	E1	US-01, US-02, US-03
MR-1 (flow from registration to billing)	S4 Send to Billing	E1	US-04
MR-2 Waiting area shows the token being served	M2 Call Next + Now-Serving display	E2	US-05, US-06
MR-2 (absent patients, accessibility)	S1 Skip/Recall, S2 Voice	E2	US-07, US-08
MR-3 Each patient sees an estimated wait	M3 Estimated waiting time	E3	US-09, US-10
MR-3 (several counters)	S5 Multiple counters	E3	US-11
MR-1 / MR-3 (patient keeps their token)	S3 QR tracking; slip from M1	E4	US-12, US-13
MR-1 / MR-2 (each role reaches its own screen)	S6 Role-based sign-in	E5	US-14
MR-3 / MR-4 (measure real waits, no extra cost)	C2 Daily stats	E6	US-15, US-16
MR-1 (patients who cannot wait)	C1 Priority tokens	E1	US-17
MR-2 (patients who read Tamil)	C3 Local-language display	E2	US-18
MR-3 (step away safely)	C4 Turn-is-near alert	E4	US-19
MR-3 (how patients feel about the wait)	C2 Daily stats: feedback	E6	US-20

User need (MRD)	PRD feature	Epic	Stories
MR-3 (honest estimates when the doctor is late)	M3 Estimated waiting time	E3	US-21
MR-2 (absent patients keep their turn)	S1 Skip / Recall	E2	US-22
MR-3 (family can follow the token)	S3 QR tracking	E4	US-23
MR-3 (estimates are measurably right)	M3 + Success metric: EWT accuracy	E6	US-24
MR-3 / MR-4 (spread the crowd)	C2 Daily stats	E6	US-25

4. Build Order

Order	What	Stories	Lead
1	Must: token issue	US-01, US-02, US-03	Dev: Ashik Rasool
2	Must: calling and display	US-05, US-06	Dev: Ashik Rasool; screens by Nishandhini
3	Must: waiting-time estimate	US-09, US-10	Dev: Ashik Rasool
4	Should, if time allows	US-07, US-08, US-04, US-13, US-11	Dev, after QA passes 1-3
5	Could	US-12	Dev
6	Should: sign-in (built by the team)	US-14	Dev: Ashik Rasool
7	Could: queue reports (added by the team)	US-15, US-16	Dev: Ashik Rasool
8	Could: priority, Tamil, alerts, feedback (added by the team)	US-17, US-18, US-19, US-20	Dev: Ashik Rasool
9	Could: delay notice, hold my place, share, accuracy, best time (added by the team)	US-21 to US-25	Dev: Ashik Rasool
Each step	Acceptance criteria tested as each story lands	All	QA: Rogasree

Scope note: doctor consultation queues, payments, medical records and remote booking are out of scope (PRD section 4) and have no stories.